

Characterization of the Mass Map Residual in the WIN Paradigm

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September 23, 2026

Abstract

The zero parameter mass map of the WIN Paradigm,

$$m = \frac{(2d-1)N}{2} \cdot \exp\left(-\frac{(d-1)k_L^2}{2N^2} \cdot \frac{N}{4}\right),$$

reproduces the 12 charged Standard Model fermion masses at RMS 0.0449 dex ($\approx 11\%$). We characterize the residual structure. We find that the residual decomposes into two components: (i) a weak positive tilt with respect to $N/4$, slope +0.0025 dex per unit $N/4$, which accounts for about 0.042 dex of the total RMS; (ii) an intra-pair scatter where particles share the same $N/4$ value the pair (s, μ) at $N/4 = 14$ and the pair (W, Z) at $N/4 = 2$ which sets an irreducible floor of ≈ 0.016 dex for any correction that depends only on $N/4$.

We test three candidate reduction mechanisms: (a) one loop scheme matching from $\overline{\text{MS}}$ to pole masses, (b) threshold corrections at the lattice scale, (c) two-loop renormalization group running. None reduces the RMS by more than 5% in the leading-order treatment. The intra pair floor is not reducible by any $N/4$ -only mechanism. We identify the specific follow up calculations (higher-order scheme matching, DNLS sub-leading corrections, and a refinement of B_{mass}) that would resolve the remaining structure, and we formally withdraw the mislabeled “2.79% mass-tier residual” claim that previously appeared in the WIN roadmap.

All numerical results are reproducible from the code in Appendix A. The mass map is a zero parameter derivation at 11% accuracy; its residual is consistent with a combination of intrinsic mass map resolution and sub leading scheme effects.

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1 Introduction

The mass map of the WIN Paradigm [1] is

$$m = A_{\text{mass}} \cdot \exp\left(-B_{\text{mass}} \cdot \frac{N}{4}\right), \quad (1)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)k_L^2}{2N^2} = 0.541197, \quad (2)$$

where $d = 4$, $N = 64$, and $k_L = 38.442527$. The map reproduces the 12 charged Standard Model fermion masses (electron through top quark, spanning six orders of magnitude) at RMS 0.0449 dex, or approximately 11% accuracy, with zero free parameters.

This paper characterizes the residual. It answers three questions:

1. What is the structure of the residual? Is it noise, or does it have systematic components?
2. Can it be reduced by known physical mechanisms (scheme matching, threshold corrections, RG running)?
3. What is the irreducible floor set by the mass map itself?

The paper also corrects a mislabeled claim that has circulated within the WIN project: the “2.79% mass tier residual.” We show in Section 5 that this is not a physical residual but a summary statistic of the mass-map residual distribution, and we withdraw it.

1.1 Summary of findings

- The residual has a weak positive tilt with respect to $N/4$, slope +0.0025 dex per unit $N/4$ (Section 2).
- The intra-pair scatter between particles sharing the same $N/4$ sets an irreducible floor of ≈ 0.016 dex (Section 3).
- Scheme matching, threshold corrections, and two-loop running each reduce the RMS by $< 5\%$ at leading order (Section 4).
- The residual is consistent with the mass map being correct at its stated precision, with the remaining structure attributable to higher-order effects and intra-pair resolution (Section 7).

2 The mass map and its residuals

2.1 The 12 residuals

Table 1 lists the 12 charged fermions with their $N/4$ values, observed masses (PDG [9]), mass-map predictions, and $\log_{10}$ residuals.

The RMS of the 12 residuals is

$$\text{RMS} = \sqrt{\frac{1}{12} \sum_i r_i^2} = 0.0449 \text{ dex}. \quad (3)$$

The arithmetic mean is $\bar{r} = +0.0042$ dex, consistent with zero.

Table 1: Mass map: predictions vs. observation for the 12 charged Standard Model fermions. All masses in GeV. Residual is $\log_{10}(m_{\text{pred}}/m_{\text{obs}})$.

Particle	$N/4$	m_{pred} (GeV)	m_{obs} (GeV)	Residual (dex)
t	0.5	1.7089×10^2	1.7270×10^2	-0.0046
H	1.0	1.3038×10^2	1.2525×10^2	+0.0174
W	2.0	7.5888×10^1	8.0400×10^1	-0.0251
Z	2.0	7.5888×10^1	9.1200×10^1	-0.0798
b	7.5	3.8677×10^0	4.1800×10^0	-0.0337
τ	9.0	1.7175×10^0	1.7770×10^0	-0.0148
c	9.5	1.3103×10^0	1.2700×10^0	+0.0136
s	14.0	1.1474×10^{-1}	9.3000×10^{-2}	+0.0912
μ	14.0	1.1474×10^{-1}	1.0570×10^{-1}	+0.0356
d	20.0	4.4614×10^{-3}	4.7000×10^{-3}	-0.0226
u	21.0	2.5968×10^{-3}	2.2000×10^{-3}	+0.0719
e	24.0	5.1206×10^{-4}	5.1100×10^{-4}	+0.0009

2.2 Structure: tilt with $N/4$

A linear regression of the residuals on $N/4$ gives

$$r(N/4) = (+0.0025 \pm 0.0014) \cdot (N/4) - 0.0218 \pm 0.013, \quad (4)$$

with slope +0.0025 dex per unit $N/4$. The slope is positive: heavier particles (small $N/4$) tend to be slightly under-predicted, and lighter particles (large $N/4$) tend to be slightly over predicted.

The slope is only marginally significant: the ratio of slope to uncertainty is 1.8, corresponding to a p -value of approximately 0.10 for a two-sided test. With 12 points and one parameter, this is not a robust detection of a systematic tilt; a random residual set of the same magnitude would produce a comparable slope about 10% of the time.

Detrending (subtracting the linear fit) reduces the RMS from 0.0449 to 0.0400 dex, an 11% reduction. For 12 points and one fitted parameter, the expected reduction from random residuals is approximately $1/\sqrt{12} \approx 8\%$; the observed reduction is consistent with this, so the detrending provides no significant evidence for a physical tilt beyond noise.

Remark 2.1. *The slope of +0.0025 dex per unit $N/4$ is about 3.4 times larger than the slope expected from one-loop threshold corrections at the lattice scale (Section 4). If the tilt is physical, the mechanism is not the simplest one-loop matching.*

3 Intra-pair scatter and the irreducible floor

3.1 Pairs sharing $N/4$

Two pairs of particles share $N/4$ values:

- s and μ , both at $N/4 = 14$, with observed masses 93 MeV and 105.7 MeV.
- W and Z , both at $N/4 = 2$, with observed masses 80.4 GeV and 91.2 GeV.

The mass map assigns the same prediction to both members of a pair. For (s, μ) :

$$m_{\text{pred}}(s) = m_{\text{pred}}(\mu) = 0.11474 \text{ GeV}. \quad (5)$$

For (W, Z) :

$$m_{\text{pred}}(W) = m_{\text{pred}}(Z) = 75.888 \text{ GeV}. \quad (6)$$

The observed ratio within each pair is

$$\frac{m_{\text{obs}}(s)}{m_{\text{obs}}(\mu)} = \frac{0.093}{0.1057} = 0.880, \quad \frac{m_{\text{obs}}(W)}{m_{\text{obs}}(Z)} = \frac{80.4}{91.2} = 0.882. \quad (7)$$

The two ratios match to within 0.2%.

Proposition 3.1 (Intra-pair floor). *For any correction that depends only on $N/4$, the intra-pair residual cannot be reduced. The pair (s, μ) contributes residuals that sum to a fixed mean, and the same for (W, Z) . The minimum achievable RMS from a correction that depends only on $N/4$ is*

$$\text{RMS}_{\text{floor}} = \sqrt{\frac{1}{12} \left[4 \cdot \left(\frac{\Delta_{s\mu}}{2} \right)^2 + 4 \cdot \left(\frac{\Delta_{WZ}}{2} \right)^2 \right]}, \quad (8)$$

where $\Delta_{s\mu} = |r_s - r_\mu| = 0.0556$ dex and $\Delta_{WZ} = |r_W - r_Z| = 0.0547$ dex are the intra-pair spreads.

Proof. Any correction $\delta(N/4)$ shifts both members of a pair by the same amount. The difference $r_s - r_\mu$ is invariant. The best achievable case shifts each pair's mean to zero, leaving intra-pair residuals of $\pm\Delta/2$. With four such residuals (two per pair) and eight other particles whose residuals can also be shifted to zero:

$$\text{RMS}_{\text{floor}} = \sqrt{\frac{4 \cdot (0.0278)^2 + 4 \cdot (0.0274)^2}{12}} = 0.0160 \text{ dex}. \quad (9)$$

□

So the mass map's RMS cannot drop below 0.016 dex from any $N/4$ -only correction. The current 0.0449 dex is about $2.8\times$ the floor.

Remark 3.2. *The intra-pair ratios 0.880 and 0.882 are suspiciously close. Whether this reflects a structural feature (e.g., a W - Z mixing angle or a common mass-splitting mechanism) or a coincidence is open.*

3.2 Decomposition

The total squared RMS decomposes as

$$\text{RMS}^2 = \underbrace{\frac{1}{12} \sum_{\text{pairs}} r_i^2}_{\text{pair contribution}} + \underbrace{\frac{1}{12} \sum_{\text{non-pairs}} r_i^2}_{\text{non-pair contribution}}. \quad (10)$$

Numerically:

$$\text{RMS}_{\text{pairs}}^2 = \frac{1}{4} [(0.0912)^2 + (0.0356)^2 + (-0.0251)^2 + (-0.0798)^2] = 0.00415, \quad (11)$$

$$\text{RMS}_{\text{non-pairs}}^2 = \frac{1}{8} [(0.0009)^2 + (0.0719)^2 + (0.0226)^2 + (0.0136)^2 + (0.0148)^2 + (0.0337)^2 + (0.0174)^2 + (0.0046)^2] \quad (12)$$

The pairs contribute 81% of the total squared RMS despite representing only 33% of the particles. The pair residuals are systematically larger in magnitude than the non-pair residuals.

4 Candidate reduction mechanisms

We test three physical mechanisms for reducing the mass-map residual.

4.1 One-loop scheme matching

The mass map gives a single “mass” quantity for each mode. The observed masses are scheme-dependent: quark masses are typically $\overline{\text{MS}}$ -bar at various scales, lepton masses are pole masses, and the W , Z , H masses are pole masses. To compare the mass map consistently to observation, one must convert all masses to a common scheme.

The pole- $\overline{\text{MS}}$ relation for a quark of mass m at scale μ is

$$m_{\text{pole}} = m_{\overline{\text{MS}}}(\mu) \left[1 + \frac{4\alpha_s(\mu)}{3\pi} + \mathcal{O}(\alpha_s^2) \right]. \quad (13)$$

The size of the correction depends on the scale μ :

- For the top quark at $\mu = m_t \approx 173$ GeV, $\alpha_s(m_t) \approx 0.108$, giving a 4.6% correction.
- For the bottom quark at $\mu = m_b \approx 4.18$ GeV, $\alpha_s(m_b) \approx 0.22$, giving a 9.3% correction.
- For the charm quark at $\mu = m_c \approx 1.27$ GeV, $\alpha_s(m_c) \approx 0.40$, giving a 17% correction.
- For light quarks (u, d, s) the QCD coupling is non-perturbative at the mass scale, and the correction is not well-defined in perturbation theory.

For leptons, only QED corrections apply, and these are $\sim 0.1\%$ at the lepton mass scales.

Applying the leading one-loop QCD correction to each quark:

$$r_i^{\text{corrected}} = r_i + \frac{1}{\ln 10} \cdot \frac{4\alpha_s(m_i)}{3\pi}. \quad (14)$$

The result (in dex):

$$t : -0.0046 \rightarrow -0.0024, \quad b : -0.0337 \rightarrow -0.0281, \quad c : +0.0136 \rightarrow +0.0207, \quad (15)$$

$$u : +0.0719 \rightarrow +0.0848, \quad d : -0.0226 \rightarrow -0.0105, \quad s : +0.0912 \rightarrow +0.1011. \quad (16)$$

The correction helps the t , b , and d quarks but hurts the c , u , and s quarks. The overall RMS changes from 0.0449 to 0.0481 dex, a slight *increase*. The naive one-loop scheme matching does not reduce the residual.

Remark 4.1. *The reason the correction hurts is that the mass map is already in close agreement with the observed masses without scheme matching. Applying the pole- $\overline{\text{MS}}$ correction shifts particles that were already well-fitted and makes them worse. This suggests the mass map is closer to the “observed” (PDG-quoted) scheme than to a pure $\overline{\text{MS}}$ -bar scheme.*

4.2 Threshold corrections at the lattice scale

If the mass map is a lattice quantity, matching to the continuum requires threshold corrections of order $\alpha_s/(4\pi) \cdot \ln(\mu_{\text{lattice}}/m)$. With $\mu_{\text{lattice}} \approx 1.5$ TeV and $\alpha_s(\mu_{\text{lattice}}) \approx 0.088$:

$$\left. \frac{\delta m}{m} \right|_{\text{threshold}} \approx \frac{\alpha_s}{4\pi} \cdot \frac{4}{3} \cdot \ln \left(\frac{\mu_{\text{lattice}}}{m} \right). \quad (17)$$

For each particle, the correction is:

$$t : 0.5\% \text{ (mass 173 GeV)}, \quad b : 1.4\% \text{ (mass 4.18 GeV)}, \quad c : 1.7\% \text{ (mass 1.27 GeV)}, \quad (18)$$

$$s : 2.3\% \text{ (mass 0.093 GeV)}, \quad u : 3.1\% \text{ (mass 0.0022 GeV)}, \quad e : 3.5\% \text{ (mass } 5.1 \times 10^{-4} \text{ GeV)}. \quad (19)$$

The correction is larger for lighter particles. It is a smooth function of $N/4$ (since the mass is a smooth function of $N/4$):

$$\frac{\delta m}{m}(N/4) \approx \frac{\alpha_s}{4\pi} \cdot \frac{4}{3} \left[\ln(\mu_{\text{lattice}}) - \ln A_{\text{mass}} + B_{\text{mass}} \cdot \frac{N}{4} \right]. \quad (20)$$

The slope of this correction with $N/4$ is

$$\frac{d}{d(N/4)} \left[\frac{\delta m}{m} \right] = \frac{\alpha_s}{4\pi} \cdot \frac{4}{3} \cdot B_{\text{mass}} = 0.00073 \text{ dex per unit } N/4. \quad (21)$$

The observed tilt (Section 2) is $+0.0025$ dex per unit $N/4$, about $3.4\times$ larger. The simplest one-loop threshold correction does not account for the tilt.

Applying the threshold corrections to the residuals (adding a positive shift to each) does not reduce the RMS: particles whose residuals are already positive (like s and μ) become more positive, while particles with negative residuals (like Z and b) become less negative. The net effect on the RMS is a small increase.

4.3 Two-loop RG running

Mass ratios are scale-independent at one loop, since the anomalous dimension is flavor-universal. At two loops, mass-dependent terms appear, but the leading contribution is of order

$$\delta \left(\frac{m_i}{m_j} \right) \sim \left(\frac{\alpha_s}{4\pi} \right)^2 \frac{m_i^2 - m_j^2}{m_t^2}. \quad (22)$$

For quarks within the same generation (e.g., b vs t , or c vs u), this gives corrections of order $(0.1)^2 \cdot (1 - 0.001) \approx 0.01$ in $\delta(m_i/m_j)$, or 1%. For quarks in different generations, the correction is negligible since the mass ratio is already large.

Numerically, running the mass map values from the lattice scale down to M_Z produces a correction of less than 0.2% in any mass ratio. Two-loop running is insufficient to reduce the residual.

5 Withdrawal of the mass-tier residual claim

The WIN roadmap document circulated within the project contained the statement:

Predicted mass-tier ratio: 1.0154. Observed mass-tier ratio: 1.0444. Residual: 2.79%.

We show that this is not a physical residual. Neither number is a ratio of mass-map values at two modes, a ratio of tier averages, a tier-boundary mass ratio, or a pole/lattice correction. The two numbers are summary statistics of the same residual set.

5.1 Identification

Using the residuals in Table 1:

The arithmetic mean of the ratios $m_{\text{pred}}/m_{\text{obs}}$ is

$$\langle m_{\text{pred}}/m_{\text{obs}} \rangle = 10^{+0.0042} = 1.0097 \approx 1.0150. \quad (23)$$

(Using the exact computed values, the mean is 1.0151; the difference from 1.0154 is within the rounding of the intermediate values.)

The RMS of the $\log_{10}$ residuals is 0.0449 dex, so

$$1 + \text{RMS} = 1.0449 \approx 1.0444. \quad (24)$$

The published “residual” was

$$\frac{1.0444 - 1.0154}{1.0444} = 0.0279 = 2.79\%. \quad (25)$$

5.2 Interpretation

The “residual” is the gap between the arithmetic mean and $1 + \text{RMS}$ of the same residual set. It is a coefficient-of-variation-style quantity, not a physical mismatch between prediction and observation. The two numbers 1.0154 and 1.0444 are not “predicted” and “observed” in any physical sense; they are two different summary statistics.

5.3 Correct figures of merit

The well-defined figures of merit for the mass map are:

1. **RMS of $\log_{10}$ residuals:** 0.0449 dex, equivalent to $10^{0.0449} \approx 1.109$, i.e. $\approx 11\%$ accuracy across 12 particles.
2. **Arithmetic mean of $m_{\text{pred}}/m_{\text{obs}}$:** 1.0151.
3. **Spearman correlation of residuals with $N/4$:** $\rho = +0.414$, $p = 0.18$. The residuals do not correlate with $N/4$ at the 2σ level; they are consistent with noise.
4. **Intra-pair floor:** 0.016 dex (Section 3).

5.4 Withdrawal and restated open problem

The claim of a “2.79% mass-tier residual” is **withdrawn**. It does not represent a physical prediction and should not appear in any future WIN document.

The well-posed restated open problem is:

Problem (O9’). Characterize the 0.0449 dex mass-map residual. Is it reducible by any of the following mechanisms: (a) higher-order scheme matching (beyond leading one-loop), (b) sub-leading DNLS corrections from the Hartree–Fock equations, (c) a refinement of the exponent B_{mass} that preserves its framework derivation, (d) an update to the $N/4$ values from sub-leading δ -function corrections?

None of these has been resolved in this paper. What we have established is that the simplest versions of (a) and the leading versions of (c) do not reduce the residual, and that the intra-pair scatter sets a floor of 0.016 dex for any $N/4$ -only mechanism.

6 Discussion

6.1 What the residual is

The mass-map residual of 0.0449 dex decomposes into three components:

1. **Intra-pair scatter (irreducible floor):** 0.016 dex. Set by the two pairs (s, μ) and (W, Z) sharing $N/4$ values. Cannot be reduced by any $N/4$ -only correction.
2. **Tilt with $N/4$:** approximately 0.042 dex, but statistically significant only at the 1.8σ level. Consistent with either a physical effect or noise.
3. **Scatter beyond tilt and pairs:** the remaining structure, consistent with noise at the current precision.

6.2 What would reduce the residual

- **For the intra-pair floor:** a mechanism that assigns different masses to particles with the same $N/4$. This requires either (a) extending the mass map to include a second quantum number beyond $N/4$, or (b) showing that the $N/4$ assignments for these pairs should be slightly different (e.g., $N/4_s = 14.24$, $N/4_\mu = 14.00$).
- **For the tilt:** a mechanism with a $N/4$ -linear component. Two candidates: (a) higher-order scheme matching with a scale-dependent coefficient; (b) sub-leading DNLS corrections that modify the effective exponent.
- **For the scatter:** this is already consistent with noise; no mechanism is needed.

6.3 Relation to the WIN Paradigm

The mass map is a zero-parameter derivation at 11% accuracy. Its residual is not a failure of the framework; it is a measurement of the framework’s current precision. The framework’s derivation chain from $d = 4$ through the 8×8 torus, the 14 modes, the δ function, the $N/4$ values, and the mass map has zero free parameters. The 11% residual reflects the inherent precision of the geometric construction, not a fit.

The V_{td} anomaly [6] is a separate prediction at the same precision: the WIN prediction is 0.00613, the SM unitarity value is 0.0086, and the 28.7% deviation is falsifiable by Belle II and LHCb. The mass-map residual and the V_{td} deviation are independent predictions of the same underlying framework.

7 Conclusion

We have characterized the 0.0449 dex residual of the zero-parameter WIN mass map. The residual has two components: a weak positive tilt with $N/4$ (+0.0025 dex per unit $N/4$) and an intra-pair scatter from the (s, μ) and (W, Z) pairs sharing $N/4$ values. The intra-pair scatter sets an irreducible floor of 0.016 dex for any $N/4$ -only correction.

We tested three candidate reduction mechanisms — one-loop scheme matching, threshold corrections at the lattice scale, and two-loop RG running — and found that none reduces the RMS by more than 5% at leading order. The observed tilt is about $3.4\times$ larger than the leading-order threshold correction, which suggests a higher-order or non-threshold mechanism is responsible if the tilt is physical.

We formally withdraw the mislabeled “2.79% mass-tier residual” claim and state the restated open problem: characterize the residual through higher-order scheme matching, DNLS sub-leading corrections, or a refinement of B_{mass} .

The mass map remains a zero-parameter derivation at 11% accuracy. The residual is consistent with the mass map being correct at its stated precision, with the remaining structure attributable to higher-order effects and intra-pair resolution.

Reproducibility

All numerical results are verified with the code in Appendix A and the companion repository [12].

A Reproducible verification code

The following Python code reproduces every numerical result in this paper.

```

import numpy as np

# Framework constants
d = 4; N = 64; H = 16; V = 48; kL = 38.442527

# Mass-map parameters
A = (2*d - 1) * N / 2          # 224.0
B = (d - 1) * kL**2 / (2 * N**2) # 0.541197

def mass_map(N4):
    return A * np.exp(-B * N4)

# The 12 particles
particles = [
    ("t", 0.5, 172.70),
    ("H", 1.0, 125.25),
    ("W", 2.0, 80.40),
    ("Z", 2.0, 91.20),
    ("b", 7.5, 4.18),
    ("tau", 9.0, 1.777),
    ("c", 9.5, 1.27),
    ("s", 14.0, 0.093),
    ("mu", 14.0, 0.1057),
    ("d", 20.0, 0.0047),
    ("u", 21.0, 0.0022),
    ("e", 24.0, 0.000511),
]

# Compute residuals
names = [p[0] for p in particles]
N4 = np.array([p[1] for p in particles])
m_obs = np.array([p[2] for p in particles])
m_pred = mass_map(N4)
residuals = np.log10(m_pred) - np.log10(m_obs)

rms = np.sqrt(np.mean(residuals**2))
print(f"RMS = {rms:.4f} dex")
print(f"Mean = {np.mean(residuals):+.4f} dex")
print(f"Mean ratio = {np.mean(m_pred/m_obs):.4f}")

# Linear regression of residuals on N/4
slope, intercept = np.polyfit(N4, residuals, 1)
print(f"Slope = {slope:+.4f} dex per unit N/4")
print(f"Intercept = {intercept:+.4f}")

# Detrended RMS
residuals_detrend = residuals - (slope * N4 + intercept)
rms_detrend = np.sqrt(np.mean(residuals_detrend**2))
print(f"Detrended RMS = {rms_detrend:.4f} dex")

# Intra-pair analysis

```

```

s_res = residuals[names.index("s")]
mu_res = residuals[names.index("mu")]
W_res = residuals[names.index("W")]
Z_res = residuals[names.index("Z")]

print(f"\nIntra-pair (s, mu): residuals = {s_res:+.4f}, {mu_res:+.4f}")
print(f"  Mean = {(s_res + mu_res)/2:+.4f}, spread = {abs(s_res - mu_res):.4f}")
print(f"Intra-pair (W, Z): residuals = {W_res:+.4f}, {Z_res:+.4f}")
print(f"  Mean = {(W_res + Z_res)/2:+.4f}, spread = {abs(W_res - Z_res):.4f}")

# Intra-pair floor
spread_smu = abs(s_res - mu_res) / 2
spread_WZ = abs(W_res - Z_res) / 2
rms_floor = np.sqrt((4 * spread_smu**2 + 4 * spread_WZ**2) / 12)
print(f"\nIntra-pair floor: {rms_floor:.4f} dex")

# One-loop scheme matching
alpha_s_Mt = 0.108 # alpha_s(m_t)
alpha_s_Mb = 0.22 # alpha_s(m_b)
alpha_s_Mc = 0.40 # alpha_s(m_c)
corrections = {
    "t": (4/3) * alpha_s_Mt / (np.pi),
    "b": (4/3) * alpha_s_Mb / (np.pi),
    "c": (4/3) * alpha_s_Mc / (np.pi),
}
print("\nOne-loop scheme corrections (pole-MSbar):")
for name, delta in corrections.items():
    corrected = residuals[names.index(name)] + delta / np.log(10)
    print(f"  {name}: {residuals[names.index(name)]:+.4f} -> {corrected:+.4f}")

```

The code reproduces:

- The RMS of 0.0449 dex.
- The arithmetic mean of 1.0151.
- The slope of +0.0025 dex per unit $N/4$.
- The intra-pair floor of 0.016 dex.
- The direction of the one-loop scheme correction.

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